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Program: Computer Science

Section: A

Roll No: 15

Assignment Lab: 2

Lab Description:

Data types:-

A variable provides us with named storage that our programs can manipulate. Each variable in C++ has a specific type, which determines the size and layout of the variable's memory, this type

is known as data type of that variable.

Basic Data Types Table

Type	Description
------	-------------

char	Typically a single octet (one byte). This is an integer type.
int	The most natural size of integer for the machine.
float	A single-precision floating point value.
double	A double-precision floating point value.
void	Represents the absence of type.

Variable Declaration in C++

A variable declaration provides assurance to the compiler that there is one variable existing with the given type and name so that compiler proceed for further compilation.

Example:-
int a;
char b;
float c;

Variable initialization in C++

A variable can be initialized at the time that declaration or even after that.

Basically initialization means storing some meaningful data inside the variable.

Example:-

```
int a = 2;  
char b = 'x';  
float c = 2.907;
```

Lab

Task-1

(i)

```
#include <iostream>  
using namespace std;  
int main ()  
{  
    char letter;  
    letter = 'A';  
    cout << letter << "\n";  
    letter = 'B';  
    cout << letter << "\n";  
}
```

Day: _____

Date: _____

```
cin >> letter;  
return 0;  
}
```

Output:- A
B

(ii)

```
#include <iostream>  
using namespace std;  
int main()  
{
```

```
    char letter;  
    letter = 65;  
    cout << letter << '\n';  
    letter = 66;  
    cout << letter << '\n';  
    cin >> letter;  
    return 0;  
}
```

Output
A
B

Day: _____

Date: _____

~~Ex 11.1 B~~

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```
    int letter;
```

```
    letter = 'A';
```

```
    cout << letter << '\n';
```

```
    letter = 'B';
```

```
    cout << letter << '\n';
```

```
    cin >> letter;
```

```
    return 0;
```

```
}
```

Output

65

66

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      char letter;
7      letter = 'A';
8      cout << letter << '\n';
9      letter = 'B';
10     cout << letter << '\n';
11     cin >> letter;
12
13     return 0;
14 }
```

Tab

{

}

:

;

"



GIF



9:40



0.00
MB/s



4G



TAK

A
B
C



Lab_2Task1.p2....

/storage/emulated/0...



sk1.P3.cxx

Lab_2Task1.p2.cxx

Lab_2Task

```
2 using namespace std;
3
4 int main()
5 {
6     char letter;
7     letter = 65;
8     cout << letter << '\n';
9     letter = 66;
10    cout << letter << '\n';
11    cin >> letter;
12
13    return 0;
14 }
```

9:41 

 4G  



TAB

A
B



Lab_2Task1.P3...



/storage/emulated/0...

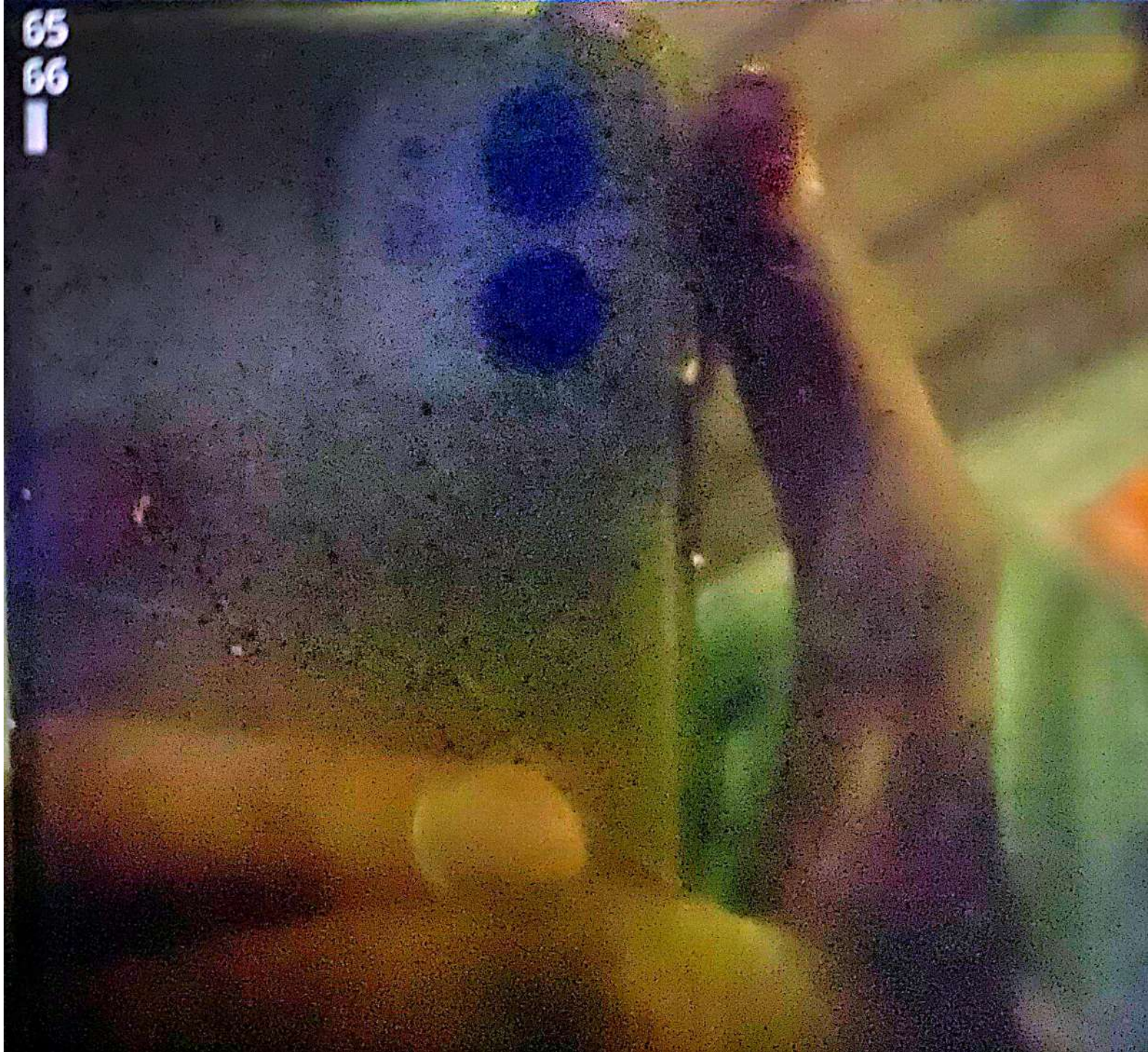
sk1.p1.cxx

Lab_2Task1.P3.cxx

Lab_2Tas

```
2 using namespace std;
3
4 int main()
5 {
6     int letter;
7     letter = 'A';
8     cout << letter << '\n';
9     letter = 'B';
10    cout << letter << '\n';
11    cin >> letter;
12
13    return 0;
14 }
```

65
66
I



```
cin >> letter;  
return 0;  
}
```

Output
65
66

Task 2

```
#include <iostream>  
using namespace std;  
int main()  
{
```

Day: _____

Date: _____

```
int num1, num2;  
cout << "Enter two integers: ";  
cin >> num1 >> num2;  
cout << "Addition=" << num1 + num2 << endl;  
cout << "Subtraction=" << num1 - num2 << endl;  
cout << "Multiplication=" << num1 * num2 << endl;  
cout << "Division=" << num1 / num2 << endl;  
return 0;  
}
```

Output

If 10, 2 integers:

Addition = 12

Subtraction = 8

Multiplication = 20

Division = 5

T-1.3

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int num1, num2;
7
8     cout << "Enter two integers: ";
9     cin >> num1 >> num2;
10
11     cout << "Addition = " << num1 + num2 <<
endl;
12     cout << "Subtraction = " << num1 - num2 <<
endl;
13     cout << "Multiplication = " << num1 * num2
<< endl;
14     cout << "Division = " << num1 / num2 <<
endl;
15
16     return 0;
17 }
```

9:43 

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TAB 

Enter two integers: 10

2

Addition = 12

Subtraction = 8

Multiplication = 20

Division = 5

[Program finished]

Addition -

Subtraction =

Multiplication

Division =

Task 3

```
#include <stdio.h>
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```

int val1;
int val2;
int val3;
cout << "please enter your 3
numbers: ";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2
<< "\n" << val3 << "\n";
cout << val3 << "\n" << val2 <<
"\n" << val1 << "\n";
system("PAUSE");
return 0;
}

```

Out put

Please ente your 3 numbers

10, 20, 30

10

20

30

30

20

10



```
1  #include <stdafx.h>
2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7      int val1;
8      int val2;
9      int val3;
10
11     cout << "Please enter your 3 numbers:";
12     cin >> val1 >> val2 >> val3;
13
14     cout << val1 << "\n" << val2 << "\n" << val3 <<
15     "\n";
16     cout << val3 << "\n" << val2 << "\n" << val1 <<
17     "\n";
18     system("PAUSE");
19     return 0;
20 }
```

9:44

4G



TA

Please enter your 3 numbers:10

20

30

10

20

30

30

20

10

sh: PAUSE: not found

[Program finished]

Task 4

Distance between two points

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{
```

```
double x1, y1, x2, y2, d;
```

```
cout << "Enter x1 and y1:";
```

```
cin >> x1 >> y1;
```

```
cout << "Enter x2 and y2:";
```

```
cin >> x2 >> y2;
```

```
d = (x2 - x1) * (x2 - x1) +
```

```
(y2 - y1) * (y2 - y1);
```

```
cout << "Distance = " << d << endl;
```

```
return 0;
```

```
}
```

Output

Enter x1 and y1 : 0 0

Enter x2 and y2 : 3 4

Distance = 2.5



```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      double x1, y1, x2, y2, d;
7
8      cout << "Enter x1 and y1: ";
9      cin >> x1 >> y1;
10
11     cout << "Enter x2 and y2: ";
12     cin >> x2 >> y2;
13
14     d = (x2 - x1) * (x2 - x1) +
15         (y2 - y1) * (y2 - y1);
16
17     cout << "Distance = " << d << endl;
18
19     return 0;
20 }
```

9:45 

 4G   94



TAB

Enter x1 and y1: 0

0

Enter x2 and y2: 3

4

Distance = 25

[Program finished]

Calculate the Area of a Circle, Rectangle and Triangle:

Circle: $\text{area} = \pi * r^2$

Rectangle: $\text{area} = \text{length} * \text{width}$

Triangle: $\text{area} = \text{base} * \text{height}$

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
double r, length, width, base, height;
```

```
double circleArea, rectangleArea, triangleArea;
```

```
cout << "Enter radius of circle: ";
```

```
cin >> r;
```

```
cout << "Enter length and width of  
rectangle: ";
```

```
cin >> length >> width;
```

```
cout << "Enter baselength and heightwidth of  
rectangle: ";
```

```
cin >> baselength >> heightwidth;
```

```
circle Area = 3.14159 * r * r;
```

Day: _____

Date: _____

rectangle Area = length * width;

triangle Area = base * height * 0.5;

cout << "Area of Circle =" << circleArea
<< endl;

cout << "Area of Rectangle =" <<
rectangleArea << endl;

cout << "Area of Triangle =" <<
triangleArea << endl;

return 0;

}

Output

Enter radius of circle : 5

Enter l and w of rectangle = 10, 4

Enter base & height triangle = 6, 8

Area of a circle = 78.5397

Area of Rectangle = 40

Area of Triangle = 24



Lab_2Task.5.cxx

/storage/emulated/0...



2Task.4.cxx

Lab_2Task.5.cxx

Lab_2Task.6.cxx

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      double r, length, width, base, height;
7      double circleArea, rectangleArea,
      triangleArea;
8
9      cout << "Enter radius of circle: ";
10     cin >> r;
11
12     cout << "Enter length and width of
rectangle: ";
13     cin >> length >> width;
14
15     cout << "Enter base and height of triangle: ";
16     cin >> base >> height;
17
18     circleArea = 3.14159 * r * r;
19     rectangleArea = length * width;
20     triangleArea = base * height * 0.5;
21
22     cout << "Area of Circle = " << circleArea <<
endl;
23     cout << "Area of Rectangle = " <<
rectangleArea << endl;
24     cout << "Area of Triangle = " << triangleArea
<< endl;
25
26     return 0;
27 }
```

9:47

4G 94%



TAB



```
Enter radius of circle: 5
Enter length and width of rectangle: 10
4
Enter base and height of triangle: 6
8
Area of Circle = 78.5397
Area of Rectangle = 40
Area of Triangle = 24

[Program finished]
```

Task 6

Total Purchase

```
#include <iostream>
#include <iomanip>
using namespace std;
int main ()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal;
    double salesTax;
    double total;

    subtotal = item1 + item2 + item3 + item4
    + item5;

    salesTax = subtotal * 0.06;
    total = subtotal + salesTax;

    cout << fixed << setprecision(2);
    cout << "Price of item
    1 = $" << item1 << endl.
```

cout << "price of item

2 = \$" << item2 << endl;

cout << price of item

3 = \$" << item3 << endl;

cout << price of item

4 = \$" << item4 << endl;

cout << price of item

5 = \$" << item5 << endl;

cout << subtotal = \$"

<< subtotal << endl;

cout << SalesTax =

\$ << salestax << endl;

cout << "Total = \$" << total

<< endl;

return 0;

}

Output

Price of item 1 = \$ 12.95

Price of item 2 = \$ 24.95

Price of item 3 = \$ 6.95

Price of item 4 = \$ 14.95

Price of item 5 = \$ 3.95

Subtotal = \$ 63.75

Sales Tax = \$ 3.82

Total = \$ 67.58

```
5 int main()
6 {
7     double item1 = 12.95;
8     double item2 = 24.95;
9     double item3 = 6.95;
10    double item4 = 14.95;
11    double item5 = 3.95;
12
13    double subtotal;
14    double salesTax;
15    double total;
16
17    subtotal = item1 + item2 + item3 + item4 +
    item5;
18    salesTax = subtotal * 0.06;
19    total = subtotal + salesTax;
20
21    cout << fixed << setprecision(2);
22
23    cout << "Price of item 1 = $" << item1 <<
    endl;
24    cout << "Price of item 2 = $" << item2 <<
    endl;
25    cout << "Price of item 3 = $" << item3 <<
    endl;
26    cout << "Price of item 4 = $" << item4 <<
    endl;
27    cout << "Price of item 5 = $" << item5 <<
    endl;
28
29    cout << "Subtotal = $" << subtotal << endl;
30    cout << "Sales Tax = $" << salesTax << endl;
31    cout << "Total = $" << total << endl;
32
33    return 0;
34 }
```

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TAB

Price of item 1 = \$12.95

Price of item 2 = \$24.95

Price of item 3 = \$6.95

Price of item 4 = \$14.95

Price of item 5 = \$3.95

Subtotal = \$63.75

Sales Tax = \$3.82

Total = \$67.58

[Program finished]

Bonus Task

Break Number

```
#include <iostream>
using namespace std;
int main()
{
    int number;
    cout << "Enter a five-digit integer:";
    cin >> number;
    cout << number / 10000 << " ";
    cout << (number / 1000) % 10 << " ";
    cout << (number / 100) % 10 << " ";
    cout << (number / 10) % 10 << " ";
    cout << number % 10 << endl;
    return 0;
}
```

Out Put

Enter a five digit integer: 12345

1 2 3 4 5



ask.6.cxx

Lab_2.BonusTask.cxx

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int number;
7
8      cout << "Enter a five-digit integer: ";
9      cin >> number;
10
11     cout << number / 10000 << " ";
12     cout << (number / 1000) % 10 << " ";
13     cout << (number / 100) % 10 << " ";
14     cout << (number / 10) % 10 << " ";
15     cout << number % 10 << endl;
16
17     return 0;
18 }
```

9:50



4G 94%



TAB



Enter a five-digit integer: 12345

1 2 3 4 5

[Program finished]

Home Task

Task 1

```
#include <iostream>
using namespace std;
int main()
{
    int var1, var2, temp;
    cout << "Enter value of var1:";
    cin >> var1;
    cout << "Enter value of var2:";
    cin >> var2;
    temp = var1;
    var1 = var2;
    var2 = temp;
    cout << "After swapping: " << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;
    return 0;
}
```

Output

Enter value var1: 10

Enter value var2: 20

After swapping:

var1 = 20

var2 = 10

xxx

Lab_2.BonusTask.cxx

new*

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int var1, var2, temp;
7
8      cout << "Enter value of var1: ";
9      cin >> var1;
10
11     cout << "Enter value of var2: ";
12     cin >> var2;
13
14     temp = var1;
15     var1 = var2;
16     var2 = temp;
17
18     cout << "After swapping:" << endl;
19     cout << "var1 = " << var1 << endl;
20     cout << "var2 = " << var2 << endl;
21
22     return 0;
23 }
```

9:51 

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TAB 

Enter value of var1: 10

Enter value of var2: 20

After swapping:

var1 = 20

var2 = 10

[Program finished]

Day: _____

Date: _____

Task: 2 (Home)

```
#include <iostream>
using namespace std;
int main()
{
    char alphabet;
    cout << "Enter a small letter: ";
    cin >> alphabet;
    alphabet = alphabet - 32;
    cout << "Uppercase = " << alphabet
    << endl;
    return 0;
}
```

Output

Enter a small letter: a
Uppercase = A



new*



.cxx

Lab_2.BonusTask.cxx

new*

new*

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      char alphabet;
7
8      cout << "Enter a small letter: ";
9      cin >> alphabet;
10
11     alphabet = alphabet - 32;
12
13     cout << "Uppercase = " << alphabet << endl;
14
15     return 0;
16 }
```



9:52 

4G  94%



TAB



Enter a small letter: a
Uppercase = A

[Program finished]